

**AI-Augmented Decision Support: A Multi-Stage System for Alert Classification,
Search, and Symbolic Reasoning**

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Introduction

In contemporary IT and cloud infrastructure landscapes, engineering teams frequently contend with a staggering influx of system alerts. These alerts exhibit a wide spectrum of severity, ranging from routine maintenance notifications to indicators of critical system failures. The manual triage of each alert is not only time-consuming but also contributes to alert fatigue, which can result in crucial notifications being inadvertently ignored. This project introduces an AI-driven decision support system aimed at enhancing alert classification and prioritization. The system integrates advanced natural language processing (NLP), supervised classification techniques, heuristic search methodologies, and symbolic reasoning to provide a multi-faceted analysis and actionable response recommendations. It ultimately bolsters human decision-making by highlighting alerts of utmost urgency and offering guidance on necessary escalation or suppression actions based on established logical frameworks.

Resources, Frameworks, and Interfaces

This project was executed in Python, leveraging well-established AI and data science libraries to enhance its functionality. For natural language processing tasks, we employed Scikit-learn, utilizing TfidfVectorizer for effective vectorization and LogisticRegression for classification purposes. Additionally, we developed a custom prioritization system based on search algorithms, utilizing the standard heapq library to efficiently manage tasks. Data manipulation and export were seamlessly handled using

Pandas, while logical rules were implemented through conditional statements, effectively simulating the behavior of an expert system. This comprehensive approach ensures both rigor and adaptability in addressing the project's objectives.

Search Techniques Employed

The program utilizes Best-First Search, a heuristic-driven methodology that assigns a priority score to each alert based on several factors, including predicted severity, system criticality, and urgency-related keywords. Alerts are organized within a priority queue and ranked through a cost function described as: $\text{priority_score} = \text{severity_weight} + \text{system_weight} + \text{keyword_bonus}$. This strategic approach ensures that high-risk alerts, such as "Database connection failed" or "Unexpected shutdown," are prioritized, allowing engineering teams to effectively triage the most critical issues first. The implementation of Best-First Search exemplifies the capacity of artificial intelligence to prioritize tasks in scenarios where a rigid rule-based ranking may be absent.

Classification and Deep Learning Techniques

A logistic regression model was developed utilizing a synthetic dataset comprising 750 alerts, classified into three distinct severity categories: critical, informational, and false positive. Each alert was enriched with a message, relevant system context, and a historical severity label. To prepare the data for machine learning, natural language processing (NLP) techniques were employed to vectorize the messages through the application of term frequency-inverse document frequency (TF-IDF), resulting in a term-document matrix that aligns with machine learning requirements. Although deep neural networks were evaluated as a potential alternative, logistic

regression was ultimately preferred due to its superior interpretability and simplicity, which are particularly advantageous when integrated with symbolic logic methodologies.

Concepts of Expert Systems and Symbolic Reasoning

The program incorporates a streamlined expert system that utilizes symbolic reasoning to categorize alert response actions. By employing first-order logic (FOL), the system thoroughly assesses alerts after classification and prioritization. It then implements decision rules such as: if the severity level is categorized as critical and the system is identified as a database or backend, the alert is escalated immediately; if the alert is deemed informational and includes terms like "maintenance" or "backup," it is suppressed; and if the alert is identified as a false positive, the ticket is automatically closed. These decision rules emulate human-like reasoning by relying on logical criteria, which align with established artificial intelligence methodologies, such as rule-based expert systems.

Exploring Knowledge Representation and Symbolic Planning

Alert data is organized in a tabular structure, where each alert is defined by a specific set of attributes, including message text, system name, predicted severity, and priority score. This configuration serves as a fundamental method for knowledge representation. Embedded within this framework is symbolic planning, manifested through the conditional logic that dictates escalation pathways—effectively creating a decision tree without requiring explicit enumeration. For instance, the decision-making process for escalating a critical alert within a database system mimics a high-priority channel intended for human intervention.

Conclusion

This project demonstrates the integration of classical AI techniques, including natural language processing (NLP), classification, search algorithms, and symbolic logic, into a unified application designed to support informed decision-making. Our AI-powered alert analyzer addresses a critical need for intelligent triage in cloud operations, serving as a prime example of hybrid AI design. Although the current implementation utilizes synthetic data, the architecture is flexible enough to accommodate enterprise alert logs. Furthermore, it can be integrated with real-time monitoring systems to facilitate effective deployment in IT environments, thereby enhancing overall operational efficiency and responsiveness.

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